PRO3, Course Assignment, Autumn 2026

Case

Your task is to simulate the operation of a nearby slaughterhouse. In one end of the slaughterhouse living animals arrive and in the other end various products, consisting of one or more animal parts, depart. The slaughterhouse has three stations where smaller parts of the whole process take place.

1. At the first station, the animals arrive, are weighed and registered.
2. At the second station, the animals are cut into smaller parts. Each part is weighed and registered, including a reference to which animal it comes from.
   The parts are put into trays with each tray containing only one type of parts. Each tray has a maximum weight capacity.
3. At the third station, products are packed for distribution. One product might be a package with a number of the same kind of parts intended for repackaging in supermarkets. Another kind of product might be "half an animal" where all the expected parts are included, but not necessarily coming from the same animal. All products are registered including references back to the trays the parts came from.

If it is later discovered, that there is some kind of trouble with a slaughtered animal, it should be possible to recall all products, which might contain parts of the animal. This function should be accessible outside the slaughterhouse.

It is important for the customer that the stations can work as independently as possible. In particular, Work shouldn't stop at a station just because the network is down.

Part 1

This part is about making the above description more precise and designing the architecture of the system. Hand in.

- A domain model of the slaughterhouse
- An architectural overview of the system. Focus on the types of communication (direct/indirect etc.), not technology. This should include the C1 (System Context View) and C2 (Container View) of the C4 architectural model. Use whatever tool/style you prefer.

Part 2

Implement a service to retrieve information about animals and products. It should be able to retrieve registration number for all animals involved in a product, as well as all products an animal has been involved in. The information must come from a database.

The service must be implemented in gRPC. It must be tested (JUnit, BloomRPC, Postman…).

Update the architectural overview of the system. This should include the C1 (System Context View), C2 (Container View), and C3 (Component View).

Part 3

Implement a registration system for the animals as a RESTful web service. The service should store date, weight, registration number and origin (i.e. farm) of the animal. It should be possible to read the details of a specific animal, all animals arriving at a particular date, and all animals of a particular origin.

The service doesn't need a graphical user interface, but it must be tested (JUnit, Postman, …).

Update the architectural overview of the system. This should include the C1 (System Context View), C2 (Container View), and C3 (Component View).